Torque-speed and efficiency report

Validation IPM 10p V-shape — Medium. Built on demand from the design's flux-linkage map and the drive limits set on the Torque-speed tab, with the same calculation the tab uses. The run's own report records the single solved operating point.

Key results

Peak torque	6.407 N m
Base speed	3343.409 rpm
Peak power	2.695 kW
Top speed reached	9000 rpm
Best efficiency	96.285%

Status

Qualified outputs pass a benchmark gate against a reference; experimental outputs have none yet.

Torque-speed envelope	Experimental: Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.
Efficiency map	Experimental: Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.
Map flux linkage, MTPA trajectory and inductances	Experimental: The d/q gates cover 12 operating points, not a map's full range of currents and angles, and do not bound the inductances.
Phase resistance and copper loss	Experimental: No wire is set, so the wire is sized to fill the slot to its limit: an estimate, not the resistance of a real winding. Set the wire for a resistance the phase-resistance gate covers.

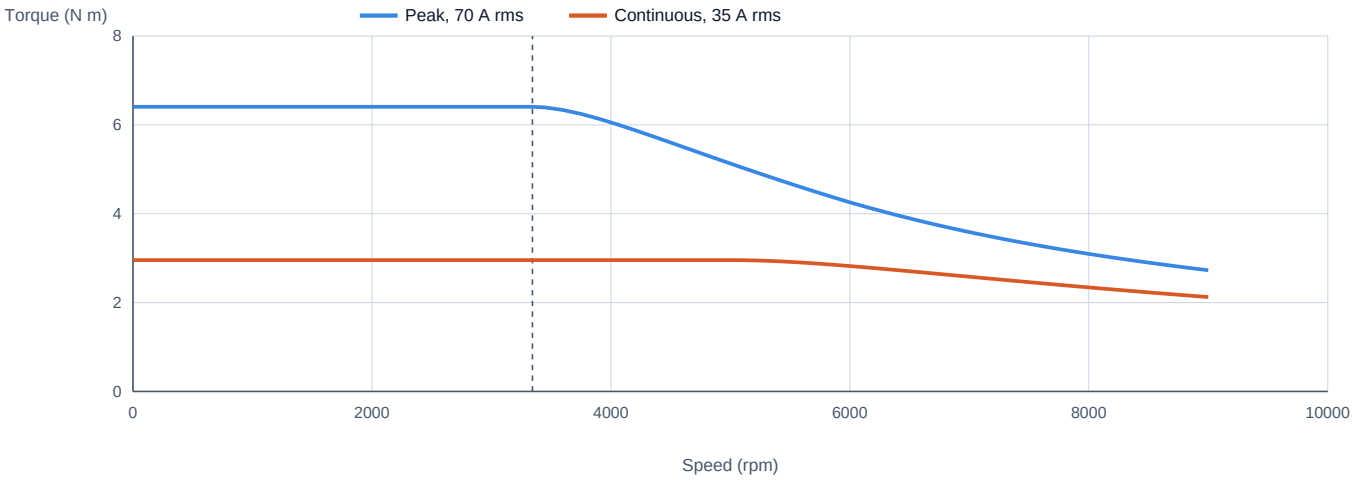
Drive inputs

Peak current	70 A rms (98.995 A peak)
Continuous current	35 A rms (49.497 A peak)
DC bus	48 V, as entered
Phase voltage limit	27.713 V peak (modulation index 1.155)
Top speed	9000 rpm
Drive control	MTPA, then field weakening (least current per torque)
Phase resistance	0.004 ohm at 20 degC, experimental
Mechanical loss	0 W
Flux-linkage map	7 currents up to 98.995 A peak x 10 angles (0 to 90 deg), 24 rotor positions each, normal mesh, standard grid; iron loss at 3000 rpm
Map scope key	539ef3f9dd6a98ed7528614c6f59c53363c82be2792eb949b9b9d3fa2bf69e91
coilEM	3ce46ed1ba1bcf19b422e504248917843dcf22cf

Torque and power against speed

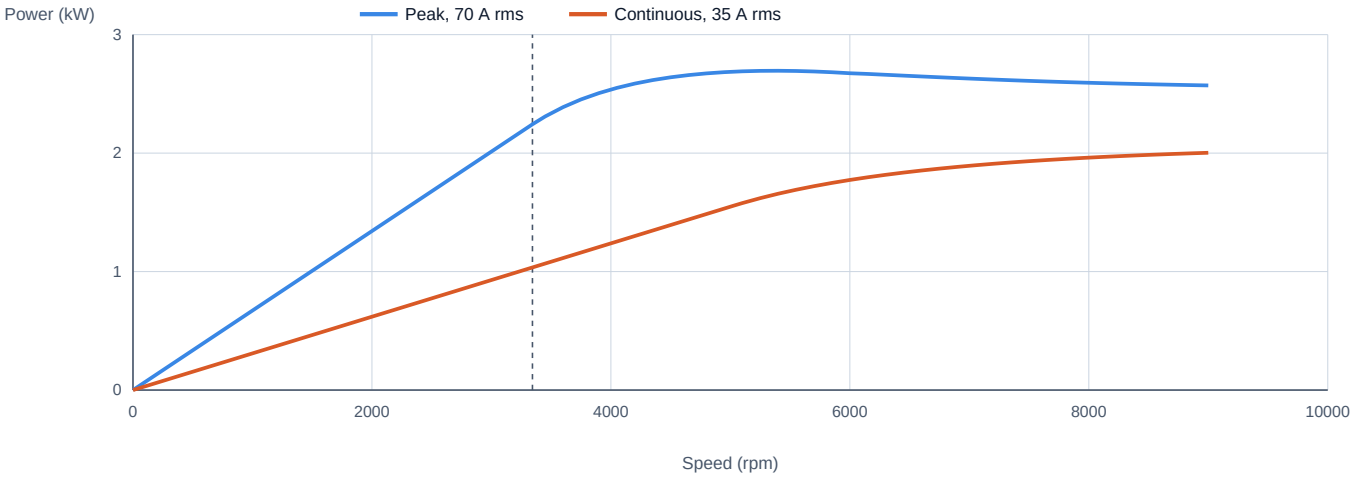
Torque against speed

Experimental



Power against speed

Experimental



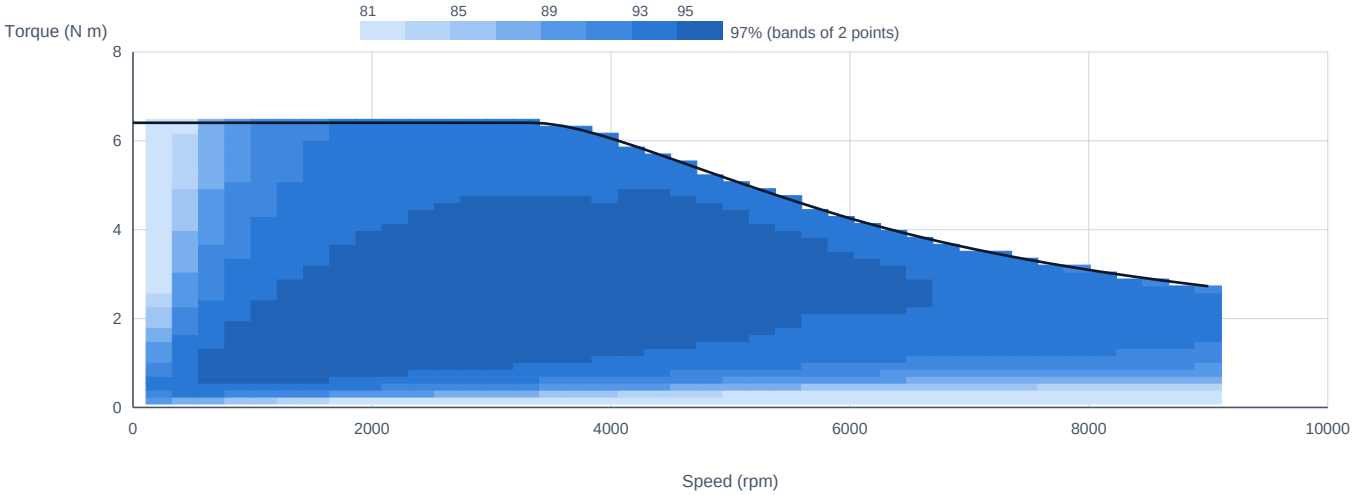
Peak curve at selected speeds (every sample is in the performance CSV)

Speed (rpm)	Torque (N m)	Power (kW)	Current (A rms)	Angle (deg)	Voltage (V peak)	Region
0	6.407	0	70	17.4	0.385	MTPA
900	6.407	0.604	70	17.4	7.621	MTPA
1650	6.407	1.107	70	17.4	13.786	MTPA
2550	6.407	1.711	70	17.4	21.188	MTPA
3300	6.407	2.214	70	17.4	27.356	MTPA
4050	6.014	2.55	70	34.617	27.713	Field weakening
4800	5.318	2.673	70	45.901	27.713	Field weakening
5700	4.505	2.689	70	55.078	27.713	Field weakening
6450	3.929	2.654	66.736	58.3	27.713	MTPV
7350	3.398	2.615	62.388	60.5	27.713	MTPV
8100	3.055	2.591	59.136	61.8	27.713	MTPV
9000	2.728	2.571	56.283	63.4	27.713	MTPV

Efficiency map

Efficiency map, MTPA, then field weakening

Experimental

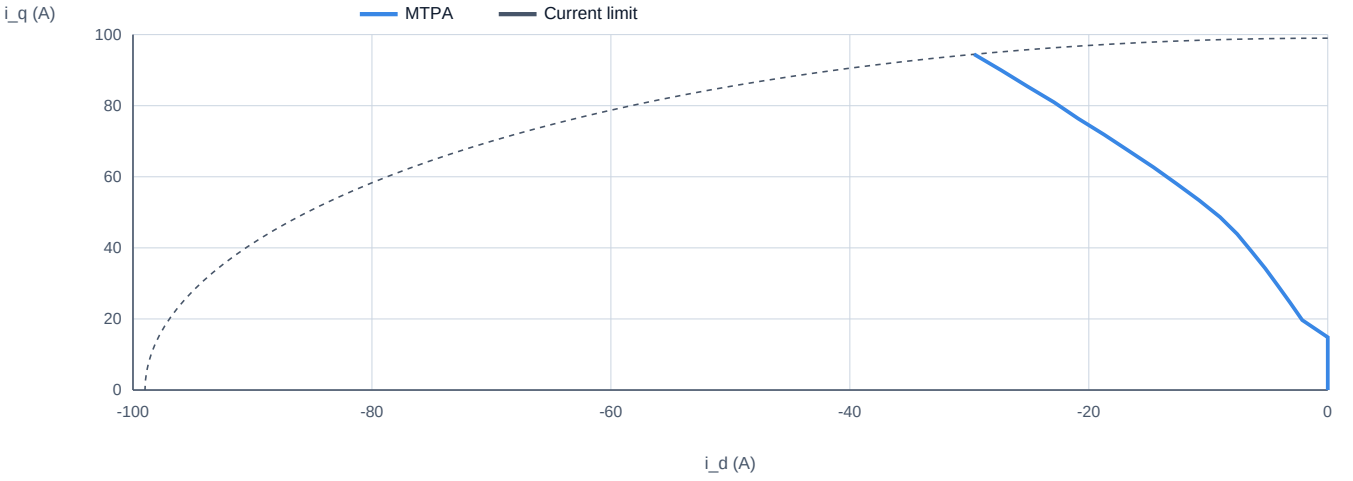


Efficiency = shaft power / input power, with shaft power = T x omega - iron loss - mechanical loss and input power = T x omega + copper loss. Cells above the peak curve are outside the drive's reach. The current and angle of every cell are in the performance CSV.

MTPA trajectory and flux linkage

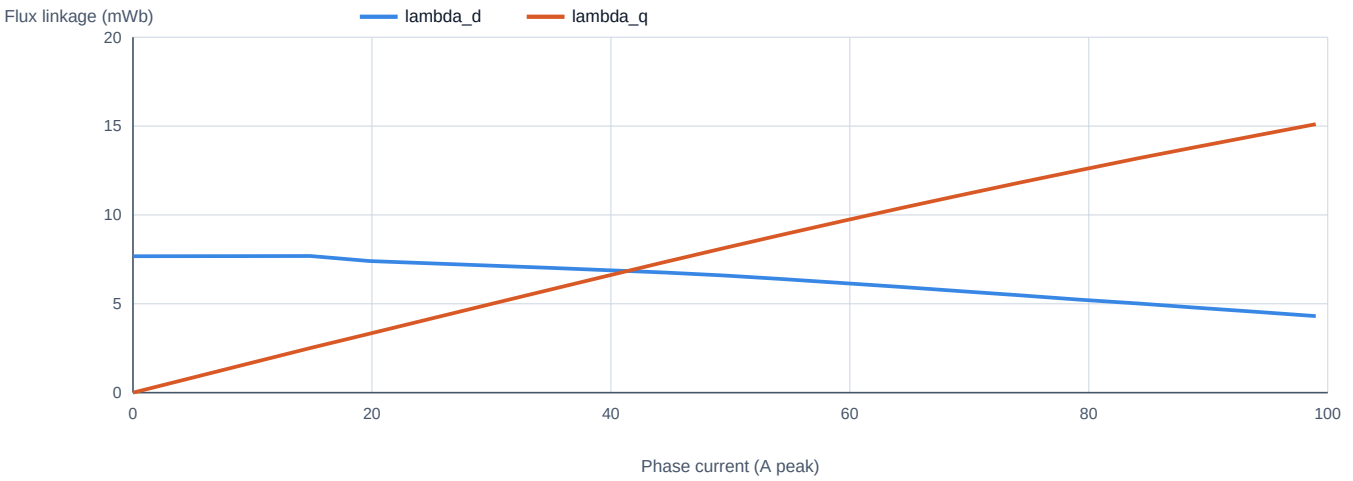
MTPA trajectory in the id-iq plane (peak A)

Experimental



Flux linkage along the MTPA trajectory

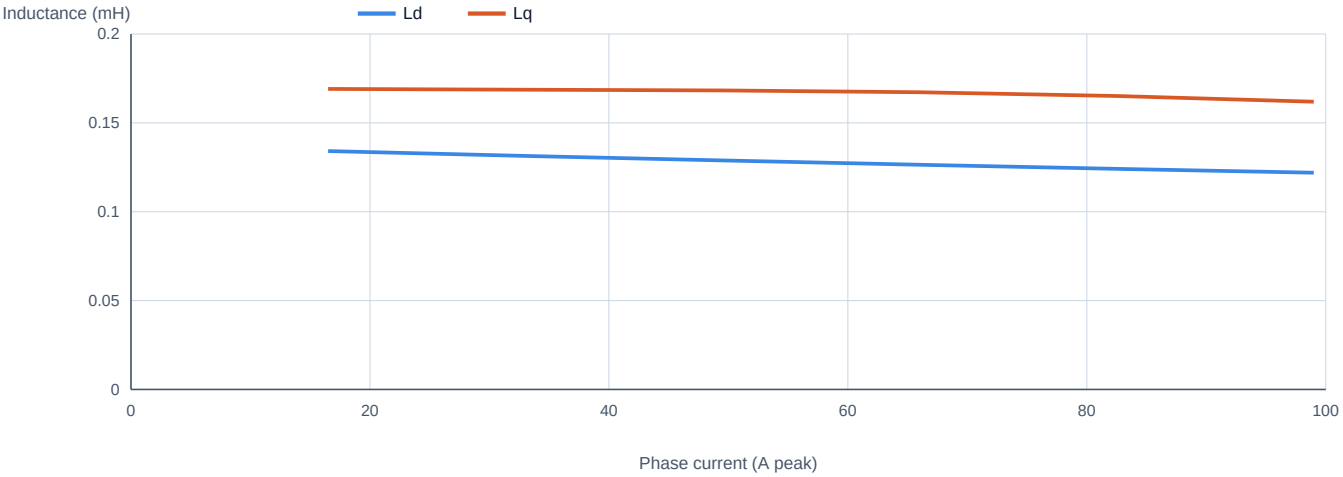
Experimental



Apparent inductance

Apparent inductance against current

Experimental



Current (A peak)	Current (A rms)	Ld (mH)	Lq (mH)
16.499	11.667	0.134	0.169
32.998	23.333	0.131	0.169
49.497	35	0.129	0.168
65.997	46.667	0.126	0.167
82.496	58.333	0.124	0.165
98.995	70	0.122	0.162

Notes

- FEM torque is 99.8-100.0% of the d/q torque across 53 map points.

- Ld from the 90 deg points, (lambda_d - lambda_pm) / i_d; Lq from the 0 deg points, lambda_q / i_q. Both are secant values and include saturation.

- The MTPA angle is read off the interpolated map, whose angles are 10 deg apart at most. At low current, where torque hardly changes with angle, it is approximate; a finer map grid gives a smoother trajectory.

- Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.

- Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.

Known limitations

- Interpolated from a flux-linkage map (Catmull-Rom over current and current angle); nothing is extrapolated beyond its currents and angles. Not yet compared with measured data.

- Steady state with sinusoidal current. Inverter voltage drops, dead time and the voltage behind the iron-loss resistance are left out.

- Torque is electromagnetic. Shaft output subtracts iron loss and the mechanical loss entered for the design.

- Iron loss is scaled from the map's reference speed: hysteresis with frequency, eddy current with its square.